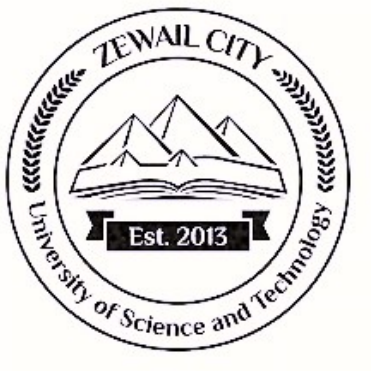
Tutorial 1: Homogeneous and Separable ODEs

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Problem 1. Find general solution to the following differential equation,

$$xy' = (1 - 4x^2)\tan y$$

Solution. The equation is separable, hence,

$$\frac{dy}{dx}\tan y = \frac{1-4x^2}{x}$$
$$\tan y dy = \left(\frac{1}{x} - 4x\right)dx$$

Integrating both sides, we get:

$$\int \tan y \, dy = \int \left(\frac{1}{x} - 4x\right)dx$$
$$\ln \sec y = \ln x - 2x^2 + c,$$
$$\ln\left(\frac{\sec y}{x}\right) = -2x^2 + c,$$
$$\sec y = c_1 x e^{-2x^2}$$
$$y = \sec^{-1}\left(c_1 x e^{-2x^2}\right)$$

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Problem 2. Find general solution to the following differential equation,

$$y - x + xy\cot x + xy' = 0$$

Solution. Rewriting the equation in the standard form,

$$\frac{dy}{dx} + \frac{1}{x}(1 + x\cot x)\, y = 1$$

the integrating factor is:

$$\mu = exp\left(\int p(x)dx\right) = exp\left(\int \left(\frac{1}{x} + \cot x\right)dx\right) = exp\left(\ln\left(x\sin x\right)\right) = x\sin x$$

Multiply the integrating factor by the whole equation:

$$x\sin x\frac{dy}{dx} + \sin x\,(1 + x\cot x)\, y = x\sin x$$
$$x\sin x\frac{dy}{dx} + (\sin x + x\cos x)\, y = x\sin x$$
$$\frac{d}{dx}(xy\sin x) = x\sin x$$

Integrating both sides with respect to x, we obtain:

$$xy\sin x = \int x\sin x \, dx$$

integrating the left right hand side by parts:

$$xy\sin x = -x\cos x + \sin x + c.$$

Rearranging the equation,

$$\boxed{y = -\cot x + \frac{1}{x} + \frac{c}{x}\csc x}$$

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Problem 3. Find general solution to the following differential equation,

$$-\frac{1}{y}\sin\frac{x}{y}dx+\frac{x}{y^2}\sin\frac{x}{y}dy=0$$

Solution. the equation in the differential form, checking if it does satisfy the test of exactness.

$$M(x,y)=-\frac{1}{y}\sin\frac{x}{y},$$
$$N(x,y)=\frac{x}{y^2}\sin\frac{x}{y}.$$

Since,

$$\frac{\partial M}{\partial y}=\frac{1}{y^2}\sin\frac{x}{y}+\frac{x}{y^3}\cos\frac{x}{y}=\frac{\partial N}{\partial x}$$

Therefore, the ODE is exact. Hence,

$$\frac{\partial f}{\partial x}=-\frac{1}{y}\sin\frac{x}{y}$$

Integrating it with respect to x:

$$f(x,y)=\cos\frac{x}{y}+g(y)$$

taking the partial derivative with respect to y:

$$\frac{\partial f}{\partial y}=\frac{x}{y^2}\sin\frac{x}{y}+g'(x)$$

also,

$$\frac{\partial f}{\partial y}=\frac{x}{y^2}\sin\frac{x}{y}$$

therefore,

$$g'(y)=0,$$
$$g(y)=c$$

Hence,

$$f(x,y)=\cos\frac{x}{y}+c=c_1$$

or, the solution is:

$$\boxed{\cos\frac{x}{y}=C}$$

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Problem 4. Find general solution to the following differential equation,

$$x^2y'=3(x^2+y^2)\tan^{-1}\left(\frac{y}{x}\right)+xy$$

Solution. Replacing each x, and y by tx, and ty, respectively, we observe that we can write the ODE as:

$$f(tx,ty)=t^2f(x,y)$$

hence, the ODE is homogeneous of degree 2. It can be transformed to separable ODE by the substitution, $z = \frac{y}{x}$.

$$\frac{dy}{dx} = \frac{3}{x^2}(x^2 + y^2)\tan^{-1}\left(\frac{y}{x}\right) + \frac{y}{x}$$
$$\frac{d}{dx}(zx) = z + x\frac{dz}{dx} = \frac{3}{x^2}(x^2 + z^2x^2)\tan^{-1}(z) + z$$

hence,

$$x\frac{dz}{dx} = 3(1 + z^2)\arctan z$$

or,

$$\frac{1}{3}\frac{1}{1+z^2}\frac{1}{\arctan z}dz = \frac{1}{x}dx$$

Integrating both sides:

$$\frac{1}{3}\ln(\arctan z) = \ln(\arctan z)^{\frac{1}{3}} = \ln x + c$$
$$\arctan z = Cx^3$$
$$z = \tan(Cx^3)$$
$$y = x\tan(Cx^3)$$

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Problem 5. Find general solution to the following differential equation,

$$y' = \frac{x^2 + xy + y^2}{x^2}$$

Solution. Doing a transformation $x \mapsto \lambda x$ and $y \mapsto \lambda y$ leaves the differential equation invariant (Check). Thus, the differential equation is homogeneous of degree zero. Thus, by choosing $\lambda = \frac{1}{x}$, we choose a transformation $y = x\nu$. Simplifying the differential equation,

$$y' = \nu + x\frac{d\nu}{dx}$$

$$\frac{x^2 + xy + y^2}{x^2} = \left(1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2\right)$$
$$= (1 + v + v^2)$$

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Removing ν from both sides,

$$x\nu' = 1 + \nu^2$$

Thus, we obtain a separable differential equation which can solved exactly,

$$\int \frac{1}{1+v^2}\frac{d\nu}{dx}dx = \int \frac{dx}{x} + c$$

which gives,

$$\tan^{-1}(\nu) = \ln(cx)$$

Producing the general solution,

$$\boxed{\tan^{-1}\left(\frac{y}{x}\right) = \ln(cx)}$$

Problem 6. Find general solution to the following differential equation,

$$y' = 2y\frac{x^2 - y}{x^3}$$

Solution. You may check that our usual transformation $x \mapsto \lambda x$ and $y \mapsto \lambda y$ would not produce a homogeneous function due to the term $(x^2 - y)$. Nevertheless, we can realize the homogeneity of the differential equation using a non-trivial transformation, $x \mapsto \lambda^m x$ and $y \mapsto \lambda^n y$. Let us check how this transformation would change the differential equation,

$$y' \mapsto \frac{d(\lambda^n y)}{d(\lambda^m x)} = \lambda^{n-m}\frac{dy}{dx}$$

$$f(x,y) = 2y\frac{x^2 - y}{x^3} = 2y\lambda^n\frac{\lambda^{2m}x^2 - \lambda^n y}{\lambda^{3m}x^3} = 2\lambda^{n-m}y\frac{x^2 - \lambda^{n-2m}y}{x^3}$$

Notice that the function would restore its homogeneity if $n = 2m$. Furthermore, we chose $m = 1$ to produce a form that can separated (other choices of m are possible however further analysis and substitution must be made). In this regard and using substitution $\lambda = \frac{1}{x}$, we chose the substitution $y = x^2\nu$. Thus,

$$y' = 2x\nu + x^2\frac{d\nu}{dx}$$

$$f(x,y) = 2y\frac{x^2}{x^2}\frac{x^2 - y}{x^3} = 2x\frac{y}{x^2}\left(1 - \frac{y}{x^2}\right) = 2x\nu(1 - \nu)$$

Thus,

$$2x\nu + x^2\frac{d\nu}{dx} = 2x\nu(1 - \nu)$$

Producing the differential equation,

$$x^2\frac{d\nu}{dx} = -2x\nu^2$$

which can be solved using separation of variables,

$$-\int\frac{d\nu}{\nu^2} = 2\int\frac{dx}{x} \implies \frac{1}{2\nu} = 2\ln(x) + C$$

which produces the general solution,

$$\boxed{e^{\frac{x^2}{2y}} = Cx^2}$$

Problem 7. Find general solution to the following differential equation,

$$(x + x^2y)y' = y - xy^2$$

Solution. We follow the same procedure we have done in problem 6 to realize the homogeneity of the differential equation. We do the transformation $x \mapsto \lambda x$ (we already choose $m = 1$) and $y \mapsto \lambda^n y$. Thus,

$$y' \mapsto \lambda^{n-1}y'$$

$$f(x,y) = \frac{y - xy^2}{x + x^2y} \mapsto \frac{\lambda^n y - \lambda^{2n+1}xy^2}{\lambda x + \lambda^{2+n}x^2y} = \lambda^{n-1}\frac{y - \lambda^{n+1}xy^2}{x + \lambda^{n+1}x^2y}$$

Thus, if $n = -1$, the differential would be homogeneous. This motivates the substitution $xy = \nu$. Thus,

$$y' = -\frac{\nu}{x^2} + \frac{1}{x}\nu'$$

$$f(x,y) = \frac{y - xy^2}{x + x^2 y} = \frac{xy - x^2y^2}{x^2 + x^3 y} = \frac{\nu(1-\nu)}{x^2(1+\nu)}$$

Thus,

$$-\frac{\nu}{x^2} + \frac{1}{x}\nu' = \frac{\nu(1-\nu)}{x^2(1+\nu)}$$

Thus,

$$\nu' = \frac{\nu(1-\nu)}{x(1+\nu)} + \frac{\nu}{x} = \frac{2\nu}{x(1+\nu)}$$

Using separation of variables,

$$\int \frac{(1+\nu)d\nu}{\nu} = \int \frac{dx}{x} \implies \ln(x) + C = \ln(\nu) + \nu$$

Which produces the general solution,

$$\boxed{\frac{C}{y} = e^{xy}}$$

Problem 8. Find general solution to the following differential equation,

$$3x^5 - y(y^2 - x^3)y' = 0$$

Solution. Notice here also that the differential equation lack homogeneous structure due to the term $y^2 - x^3$. However, an ordinary transformation with $m = 1$ would not do the trick. Thus, we make transformation an leave m arbitrary,

$$y' \mapsto \lambda^{n-m} y'$$

$$f(x,y) = \frac{3x^5}{y(y^2 - x^3)} \mapsto \lambda^{3(5m-n)} \frac{3x^5}{y(y^2 - \lambda^{3m-2n}x^3)}$$

Thus, if $3m - 2n = 0$. The DE would be homogeneous. Our choice this time would be nontrivial to produce integer n and m for simplicity. Thus, at least $m = 2, n = 3$ are needed. Thus, a adequate transformation would be,

$$y \mapsto \lambda^3 y, \qquad x \mapsto \lambda^2 x$$

However, to produce a separable equation in this transformation we need to choose $\lambda = x^{-1/2}$. You can follow this and show that it does indeed produce separable equation, however, we would end up with a hard integration. Instead, we develop another intuition behind homogeneity. If we do transformation, $\nu = y^2$ and $u = x^3$, we would restore the homogeneity of the DE. Thus,

$$y' = \frac{dy/du}{dx/du} = \frac{y_u + y_\nu \frac{d\nu}{du}}{x_u + x_\nu \frac{d\nu}{du}} = \frac{\pm\frac{1}{2}\nu^{-1/2}\frac{d\nu}{du}}{\frac{1}{3}u^{-2/3}} = \pm\frac{3}{2}\frac{u^{2/3}}{\nu^{1/2}}\frac{d\nu}{du}$$

$$f(x,y) = \frac{3x^5}{y(y^2 - x^3)} = \pm\frac{3u^{5/3}}{\nu^{1/2}(\nu - u)}$$

Thus,

$$\pm\frac{3}{2}\frac{u^{2/3}}{\nu^{1/2}}\frac{d\nu}{du} = \pm\frac{3u^{5/3}}{\nu^{1/2}(\nu-u)} \implies \nu' = \frac{1}{2}\frac{u}{(\nu-u)}$$

This equation is now homogeneous DE. Using substitution $\nu = uk$.

$$v' = k + uk'$$

$$\frac{1}{2}\frac{u}{(\nu-u)} = \frac{1}{2}\frac{1}{(\nu/u-1)} = \frac{1}{2(k-1)}$$

Thus,

$$uk' = \frac{1}{2(k-1)} - k = \frac{1/2 - k^2 + k}{k-1}$$

Which can be easily integrated by parts,

$$\int \frac{k-1}{1/2-k^2+k}dk = \int \frac{du}{u}$$

Since,

$$\frac{k-1}{1/2-k^2+k} = -\frac{1}{2}\left[\frac{2k-1}{k^2-k-1/2} - \frac{1}{k^2-k-1/2}\right] = -\frac{1}{2}\left[\frac{2k-1}{k^2-k-1/2} - \frac{4}{3}\frac{1}{(2k/\sqrt{3}-1/\sqrt{3})^2-1}\right]$$

Now, we can easily integrate the two parts,

$$\int \frac{2k-1}{k^2-k-1/2}dk = \ln(k^2-k-1/2)$$

$$\begin{aligned}
\int \frac{dk}{(2k/\sqrt{3}-1/\sqrt{3})^2-1} &= \frac{\sqrt{3}}{2}\int \frac{du}{u^2-1} \\
&= \frac{\sqrt{3}}{2}\int \frac{du}{(u-1)(u+1)} \\
&= \frac{\sqrt{3}}{4}\int du\left[\frac{1}{u-1} - \frac{1}{u+1}\right] \\
&= \frac{\sqrt{3}}{4}(\ln(u-1) - \ln(u+1)) \\
&= \frac{\sqrt{3}}{4}\ln\left(\frac{u-1}{u+1}\right) = \frac{\sqrt{3}}{4}\ln\left(\frac{2k/\sqrt{3}-1/\sqrt{3}-1}{2k/\sqrt{3}-1/\sqrt{3}+1}\right) \qquad \blacksquare
\end{aligned}$$

Thus, we would end up with,

$$-\frac{1}{2}\left[\ln(k^2-k-1/2) - \ln\left(\frac{2k/\sqrt{3}-1/\sqrt{3}-1}{2k/\sqrt{3}-1/\sqrt{3}+1}\right)^{\frac{1}{\sqrt{3}}}\right] = \ln(u) + C$$

Which can be simplified to,

$$\ln\left[(k^2-k-1/2)\left(\frac{2k/\sqrt{3}-1/\sqrt{3}+1}{2k/\sqrt{3}-1/\sqrt{3}-1}\right)^{\frac{1}{\sqrt{3}}}\right] = \ln(Cu^{-2})$$

Substituting back the value of k,

$$\left[((\nu/u)^2-(\nu/u)-1/2)\left(\frac{2(\nu/u)/\sqrt{3}-1/\sqrt{3}+1}{2(\nu/u)/\sqrt{3}-1/\sqrt{3}-1}\right)^{\frac{1}{\sqrt{3}}}\right] = Cu^{-2}$$

and lastly substituting back the value of u and ν would produce the general solution,

$$\boxed{\left[((y^2/x^3)^2-(y^2/x^3)-1/2)\left(\frac{2(y^2/x^3)/\sqrt{3}-1/\sqrt{3}+1}{2(y^2/x^3)/\sqrt{3}-1/\sqrt{3}-1}\right)^{\frac{1}{\sqrt{3}}}\right]=Cx^{-6}}$$